
PRACTICE EXAM 2

NAME: SOLUTIONS

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Consider the matrix $A = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$. Solve the following questions concerning A :

(a) Find A^{-1} .

$$\left[\begin{array}{cc|cc} 1 & 6 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 6 & 1 & 0 \\ 0 & -9 & -2 & 1 \end{array} \right] \xrightarrow{-\frac{1}{9}R_2} \left[\begin{array}{cc|cc} 1 & 6 & 1 & 0 \\ 0 & 1 & \frac{2}{9} & -\frac{1}{9} \end{array} \right] \xrightarrow{R_1 - 6R_2} \underbrace{\left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{3} & \frac{2}{3} \end{array} \right]}_{=I_2} \underbrace{\left[\begin{array}{cc|cc} 0 & 1 & \frac{2}{9} & -\frac{1}{9} \end{array} \right]}_{=A^{-1}}$$

Thus, $A^{-1} = \frac{1}{9} \begin{bmatrix} -3 & 6 \\ 2 & -1 \end{bmatrix}$

(b) Express A as a product of elementary matrices.

From part A we saw that $A^{-1} = E_{R_1 - 6R_2} E_{-\frac{1}{9}R_2} E_{R_2 - 2R_1}$. Taking the inverse, we get

$$A = (E_{R_1 - 6R_2} E_{-\frac{1}{9}R_2} E_{R_2 - 2R_1})^{-1} = \boxed{E_{R_2 + 2R_1} E_{-9R_2} E_{R_1 + 6R_2}}.$$

More explicitly:

$$\boxed{\begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}}$$

NOTE: There are more ways to factor A as a product of elementary matrices.

Problem 2. Answer the following questions regarding linear transformations.

(a) State precisely what it means for a transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be linear.

The transformation T is linear if it satisfies the following conditions

1) $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$ for all $\vec{x}, \vec{y}$ in $\mathbb{R}^n$.

2) $T(c\vec{x}) = cT(\vec{x})$ for all $\vec{x}$ in $\mathbb{R}^n$ and all c in $\mathbb{R}$.

(b) Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the matrix transformation induced by the $m \times n$ matrix A . Show that T is a linear transformation.

1) If $\vec{x}$ and $\vec{y}$ are arbitrary vectors in $\mathbb{R}^n$, we have

$$T(\vec{x} + \vec{y}) = A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} = T(\vec{x}) + T(\vec{y}). \quad \checkmark$$

2) If $\vec{x}$ is in $\mathbb{R}^n$ and c is in $\mathbb{R}$, then

$$T(c\vec{x}) = A(c\vec{x}) = c(A\vec{x}) = cT(\vec{x}). \quad \checkmark$$

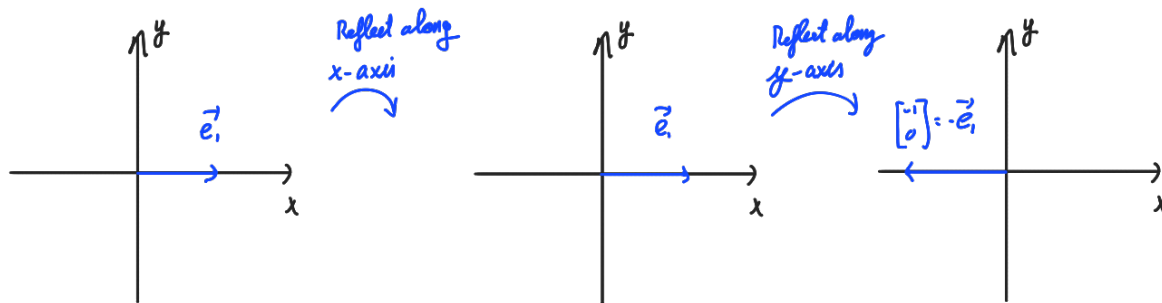
Thus, T is linear.

- (c) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects every vector $\mathbf{x}$ along the x -axis, and then reflects the result along the y -axis. Find the matrix of T and compute the image of $(1, 2)$ under T .

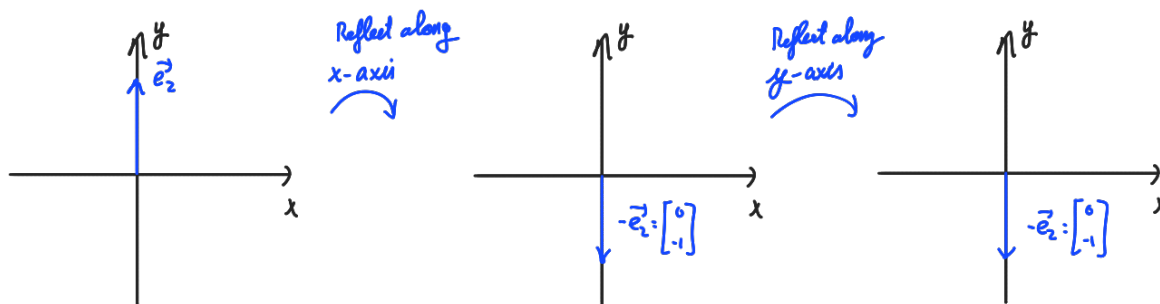
(Hint. Draw some pictures to understand the effect of the transformation T)

The matrix of T is $[T(\vec{e}_1) \ T(\vec{e}_2)]$, where $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

• Finding $T(\vec{e}_1)$:



• Finding $T(\vec{e}_2)$:



So the matrix of T is $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$.

Since $T = T_A$, the image of $(1, 2)$ under T is

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

Problem 3. (a) Using determinants, find the values of c for which the matrix

$$A = \begin{bmatrix} c & 1 & 1 \\ 1 & -c & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

is **not** invertible.

A is not invertible if and only if $\det(A) = 0$. Let's calculate $\det(A)$:

$$\begin{vmatrix} c & 1 & 1 \\ 1 & -c & 1 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow[\substack{R_2 - R_1 \\ R_3 - R_1}]{=} \begin{vmatrix} c & 1 & 1 \\ 1-c & -c-1 & 0 \\ 1-c & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1-c & -(1+c) \\ 1-c & 0 \end{vmatrix} = (1-c) \begin{vmatrix} 1 & -(1+c) \\ 1 & 0 \end{vmatrix} = (1-c)(1+c).$$

So A is not invertible if and only if $c=1$ or $c=-1$

(b) Calculate the determinant of

$$B = \begin{bmatrix} -3 & 0 & 1 & -3 & -2 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 3 & 1 \\ 0 & 0 & 7 & 2 & 0 \end{bmatrix}.$$

$$\det(B) = \begin{vmatrix} -3 & 0 & 1 & -3 & -2 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 3 & 1 \\ 0 & 0 & 7 & 2 & 0 \end{vmatrix} = -3 \begin{vmatrix} 2 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 3 & 1 \\ 0 & 7 & 2 & 0 \end{vmatrix} = (-3) \cdot 2 \cdot \begin{vmatrix} 1 & 0 & 0 \\ 3 & 3 & 1 \\ 7 & 2 & 0 \end{vmatrix}$$

$$= (-3) \cdot 2 \cdot 1 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix} = (-3) \cdot 2 \cdot 1 \cdot (-2) = \boxed{12}$$

Problem 4. On each of the following questions, **completely fill in** the circle next to the correct answer. Each question only has one correct answer. You do not need to justify your choices.

- (a) If A is a 3×3 matrix and $E_{R_1-2R_3}$ is the 3×3 elementary matrix associated with the row operation $R_1 - 2R_3$, then the result of applying the row operation $R_1 - 2R_3$ to A is:

- ☐ $AE_{R_1-2R_3}$.
☒ $E_{R_1-2R_3}A$.
☐ $E_{R_1-2R_3}^{-1}A$.
☐ $AE_{R_1-2R_3}^{-1}$.

- (b) Coming back to the matrix $E_{R_1-2R_3}$, its determinant is equal to

- ☒ $\det(E_{R_1-2R_3}) = 1$.
☐ $\det(E_{R_1-2R_3}) = -1$.
☐ $\det(E_{R_1-2R_3}) = 2$.
☐ $\det(E_{R_1-2R_3}) = -2$.

- (c) If A and B are $n \times n$ invertible matrices, then

- ☐ AB^T is invertible and its inverse is $A^{-1}(B^{-1})^T$.
☒ AB^T is invertible and its inverse is $(B^{-1})^T A^{-1}$.
☐ AB^T is invertible and its inverse is BA^T .
☐ AB^T is not invertible.

- (d) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = 3\sqrt{x} + y^3$ is linear.

- ☐ True.
☒ False.

- (e) If A is an $n \times n$ matrix, which of these conditions implies that A is invertible?

- ☐ The system $A\mathbf{x} = \mathbf{0}$ has two different solutions.
☒ $A^3 = I_n$.
☐ There exists a nonzero matrix B such that $AB = 0$.
☐ A has two equal rows.

Scratch working page